Continuum Mechanics
class # 15
10/08/2026

Change of Variables

Result (4.8): Let $\vec{\phi} : B \times [0,\infty) \to \mathbb{E}^3$ be a motion. For every $t \in [0,\infty)$ set

$$B_t = \vec{\phi}(B,t) = \vec{\phi}_t(B).$$

Assume $q : B_t \times [0,t) \to \mathbb{R}$ is an arbitrary spatial field on B_t. Suppose $\Omega_t \subseteq B_t$ with

$$\Omega_t = \vec{\phi}(\Omega,t) = \vec{\phi}_t(B).$$

Then

$$\int_{\Omega_t} q(\vec{x},t)\, dV_x = \int_{\Omega} q_m(\vec{X},t)\, \det\left(F(\vec{X},t)\right) dV_X,$$

where

$$F(X,t) = \nabla^X \vec{\phi}(X,t).$$

Proof: Any good book on Advanced Calculus. ///

Definition: Let $\vec{\phi} : B \times [0,\infty) \to \mathbb{E}^3$ be a motion. The object

$$J(\vec{X},t) := \det\left(F(\vec{X},t)\right)$$

is called the Jacobian field for the deformation $\vec{\phi}_t$.

Corollary: Suppose for a motion $\vec{\phi}: B \times [0,t) \to \mathbb{E}^3$

$$J(\vec{X}, t_0) = 1, \quad \forall \ \vec{X} \in \Omega, \quad \exists\, t_0 \in (0, \infty).$$

Then

$$Vol(\Omega) = Vol(\Omega_{t_0}).$$

Proof: Recall

$$Vol(\Omega) = \int_{\Omega} dV_X$$

and

$$Vol(\Omega_{t_0}) = \int_{\Omega_{t_0}} dV_x$$

$$= \int_{\Omega} J(\vec{X}, t_0)\, dV_X$$

$$= \int_{\Omega} dV_X$$

$$= Vol(\Omega). \quad \text{///}$$

Result (4.9): Suppose $\vec{\phi}: B \times [0, \infty) \to \mathbb{E}^3$ is a motion with spatial velocity field $\vec{v}(\vec{x}, t)$ and deformation gradient $F(\vec{X}, t)$. Then

$$\frac{\partial}{\partial t}\left(\det(F(\vec{X}, t))\right) = \det(F(\vec{X}, t))\left[\nabla^x \cdot \vec{v}\right]_m(\vec{X}, t).$$

Proof: Result (2.23) guarantees that

$$\frac{\partial}{\partial t}\left(\det\left(F(\vec{X},t)\right)\right) = \det\left(F(\vec{X},t)\right) F(\vec{X},t)^{-T} : \frac{\partial}{\partial t} F(\vec{X},t)$$

$$= \det\left(F(\vec{X},t)\right) \mathrm{tr}\left(F(\vec{X},t)^{-1} \frac{\partial}{\partial t} F(\vec{X},t)\right)$$

(4.22)
$$= \det\left(F(\vec{X},t)\right) \mathrm{tr}\left(\frac{\partial}{\partial t} F(\vec{X},t)\, F(\vec{X},t)^{-1}\right)$$

Recall that the spatial velocity satisfies

$$\vec{V}(\vec{x},t) = [\vec{v}]_m(\vec{x},t) = \frac{\partial}{\partial t}\vec{\phi}(\vec{X},t)$$

In components form

$$V_i(\vec{x},t) = \left[[\vec{v}]_m\right]_i(\vec{x},t) = \frac{\partial}{\partial t}\phi_i(\vec{X},t).$$

Next, recall that

$$F_{ij}(\vec{X},t) = \frac{\partial}{\partial X_j}\phi_i(\vec{X},t)$$

and

$$\frac{\partial}{\partial t}\frac{\partial}{\partial X_j}\phi_i(\vec{X},t) = \frac{\partial}{\partial X_j} V_i(\vec{x},t)$$

$$= \frac{\partial}{\partial X_j} v_i\left(\vec{\phi}(\vec{X},t),t\right).$$

Now, let us use the chain rule:

$$\frac{\partial}{\partial t}\frac{\partial}{\partial X_j}\phi_i(\vec{X},t) = \frac{\partial}{\partial X_j} v_i\left(\vec{\phi}(\vec{X},t),t\right)$$

$$\overset{(CR)}{=} \frac{\partial}{\partial x_k} v_i\left(\vec{\phi}(\vec{X},t),t\right) \frac{\partial \phi_k}{\partial X_j}(\vec{X},t)$$

$$= \frac{\partial}{\partial x_k} v_i\left(\vec{\phi}(\vec{X},t),t\right) F_{kj}(\vec{X},t).$$

Thus,

$$\frac{\partial F_{ij}}{\partial t}(\vec{X},t) = \frac{\partial}{\partial x_k} v_i\left(\vec{\phi}(\vec{X},t),t\right) F_{kj}(\vec{X},t),$$

or, in tensor form,

$$\frac{\partial}{\partial t} F(\vec{X},t) = \left[\nabla^x \vec{v}\right]_m (\vec{X},t)\, F(\vec{X},t).$$

Hence,

$$\left[\frac{\partial}{\partial t} F(\vec{X},t)\right] F(\vec{X},t)^{-1} = \left[\nabla^x \vec{v}\right]_m (\vec{X},t).$$

Taking the trace, we have

$$\mathrm{tr}\left[\left[\frac{\partial}{\partial t} F(\vec{X},t)\right] F(\vec{X},t)^{-1}\right] = \left[\nabla^x \cdot \vec{v}\right]_m (\vec{X},t).$$

Substituting the last expression into (4.22) gives the result:

$$\frac{\partial}{\partial t}\left(\det\left(F(\vec{X},t)\right)\right) = \det\left(F(\vec{X},t)\right)\left[\nabla^x \cdot \vec{v}\right]_m (\vec{X},t). \quad ///$$

Reynolds Transport Theorem (RTT)

Result (4.10): Let $\vec{\phi}: B \times [0,\infty) \to \mathbb{E}^3$ be a motion with associated spatial velocity $\vec{v}(\vec{x},t)$. Let $\Omega_t \subseteq B_t$ be arbitrary and denote its surface by $\partial\Omega_t$. Assume $\partial\Omega_t$ is smooth and has unit outward-pointing normal $\vec{n}$.

If $q : B_t \times [0,\infty) \to \mathbb{R}$ is an arbitrary spatial scalar field, then

$$\frac{d}{dt}\int_{\Omega_t} q(\vec{x},t)\,dV_x = \int_{\Omega_t}\left\{\dot{q}(\vec{x},t) + q(\vec{x},t)\,\nabla^x\cdot\vec{v}(\vec{x},t)\right\}dV_x,$$

or, equivalently,

(4.23) $$\frac{d}{dt}\int_{\Omega_t} q(\vec{x},t)\,dV_x = \int_{\Omega_t}\frac{\partial}{\partial t}q(\vec{x},t)\,dV_x + \int_{\partial\Omega_t} q(\vec{x},t)\,\vec{v}\cdot\vec{n}(\vec{x},t)\,dA_x.$$

Proof: By Result (4.8),

$$\int_{\Omega_t} q(\vec{x},t)\,dV_x = \int_{\Omega} q_m(\vec{X},t)\,J(\vec{X},t)\,dV_X$$

Thus,

$$\frac{d}{dt}\int_{\Omega_t} q(\vec{x},t)\,dV_x = \frac{d}{dt}\int_{\Omega} q_m(\vec{X},t)\,J(\vec{X},t)\,dV_X$$

$$= \int_{\Omega}\frac{\partial}{\partial t}\left[q_m(\vec{X},t)\,J(\vec{X},t)\right]dV_X$$

$$= \int_{\Omega}\left\{\frac{\partial}{\partial t}q_m(\vec{X},t)\,J(\vec{X},t) + q_m(\vec{X},t)\,\frac{\partial}{\partial t}J(X,t)\right\}dV_X.$$

Using the fact that

$$\left[\frac{\partial}{\partial t} q_m\right]_s (\vec{x}, t) = \dot{q}(x, t)$$

and Result (4.8) we have

$$\int_{\Omega} \frac{\partial}{\partial t} q_m(\vec{X}, t)\, J(\vec{X}, t]\, dV_X = \int_{\Omega_t} \dot{q}(\vec{x}, t)\, dV_x$$

Using Result (4.9) and Result (4.8) again,

$$\int_{\Omega} q_m(\vec{X}, t)\, \frac{\partial}{\partial t} J(X, t)\, dV_X$$

$$= \int_{\Omega} q_m(\vec{X}, t) \left[\nabla^x \cdot \vec{v}\right]_m (\vec{X}, t)\, J(\vec{X}, t)\, dV_X$$

$$= \int_{\Omega_t} q(\vec{x}, t)\, \nabla^x \cdot \vec{v}(\vec{x}, t)\, dV_x .$$

Hence,

$$\frac{d}{dt} \int_{\Omega_t} q(\vec{x}, t)\, dV_x = \int_{\Omega} \left\{ \frac{\partial}{\partial t} q_m(\vec{X}, t)\, J(\vec{X}, t] + q_m(\vec{X}, t)\, \frac{\partial}{\partial t} J(X, t) \right\} dV_X$$

$$= \int_{\Omega_t} \dot{q}(\vec{x}, t)\, dV_x + \int_{\Omega_t} q(\vec{x}, t)\, \nabla^x \cdot \vec{v}(\vec{x}, t)\, dV_x$$

$$= \int_{\Omega_t} \left\{ \dot{q}(\vec{x},t) + q(\vec{x},t) \nabla^x \cdot \vec{v}(\vec{x},t) \right\} dV_x .$$

This proves the first result. For the second, using Result (4.7), the total time derivative is

$$\dot{q}(\vec{x},t) = \frac{\partial}{\partial t} q(\vec{x},t) + \nabla^x q(\vec{x},t) \cdot \vec{v}(\vec{x},t) .$$

Thus,

$$\frac{d}{dt} \int_{\Omega_t} q(\vec{x},t)\, dV_x = \int_{\Omega_t} \left\{ \frac{\partial}{\partial t} q + \nabla^x \cdot (q \vec{v}) \right\} dV_x$$

$$= \int_{\Omega_t} \frac{\partial}{\partial t} q \, dV_x$$

$$+ \int_{\partial\Omega_t} q \vec{v} \cdot \vec{n} \, dV_x ,$$

which is the desired result. ///

Boundary Integrals

Result (4.11): Let $q(\vec{x},t)$ be an arbitrary spatial scalar field, $\vec{w}(\vec{x},t)$ an arbitrary spatial vector field, and $T(\vec{x},t)$ an arbitrary spatial tensor field.

Consider an arbitrary set $\Omega \subseteq B$ which maps to

$\Omega_t \subseteq B_t$ under the motion $\vec{\phi}_t(\vec{x})$. Let $\vec{N}(x)$ be the outward-pointing unit normal on $\partial\Omega$ and $\vec{n}(\vec{x})$ the outward-pointing unit normal on $\partial\Omega_t$. Then

$$\int_{\partial\Omega_t} q(x,t)\,\vec{n}(\vec{x})\,dA_x = \int_{\partial\Omega} q_m(\vec{X},t)\,G(\vec{X},t)\,\vec{N}(\vec{X},t)\,dA_X,$$

$$\int_{\partial\Omega_t} \vec{\omega}(\vec{x},t)\cdot\vec{n}(\vec{x})\,dA_x = \int_{\partial\Omega} \vec{\omega}_m(\vec{X},t)\cdot\left(G(\vec{X},t)\,\vec{N}(\vec{X},t)\right)dA_X,$$

$$\int_{\partial\Omega_t} T(\vec{x},t)\,\vec{n}(\vec{x})\,dA_x = \int_{\partial\Omega} T_m(\vec{X},t)\,G(\vec{X},t)\,\vec{N}(\vec{X},t)\,dA_X.$$

Proof: Any good book on Advanced Calculus. ///

Volume-Preserving Motions

Definition: A motion $\vec{\phi}: B\times[0,\infty)\to\mathbb{E}^3$ is called volume-preserving or isochoric iff

$$\text{vol}[\Omega_t] = \text{vol}[\Omega]$$

for all $t\geq 0$ and any $\Omega\subseteq B$. Here, as usual,

$$\Omega_t = \vec{\phi}_t(\Omega).$$

Result (4.12): Let $\vec{\phi}: B \times [0,\infty) \longrightarrow \mathbb{E}^3$ be a motion with spatial velocity $\vec{v}$ and deformation gradient

$$F(\vec{X},t) = \nabla^X \vec{\phi}(\vec{X},t).$$

Then $\vec{\phi}$ is volume-preserving iff

$$J(\vec{X},t) = \det(F(\vec{X},t)) = 1,$$

for all $\vec{X} \in B$ and $t \geq 0$, or, equivalently,

$$(\nabla^x \cdot \vec{v})(\vec{x},t) = 0, \quad \forall \vec{x} \in B_t, \quad \forall t \geq 0.$$

Proof: This result follows from the observations that

$$\begin{aligned} \text{vol}[\Omega_t] &= \int_{\Omega_t} dV_{\vec{x}} \\ &= \int_{\Omega} J(\vec{X},t)\, dV_X \end{aligned}$$

and

$$\frac{d}{dt}\text{vol}[\Omega_t] \overset{(RTT)}{=} \int_{\Omega_t} (\nabla^x \cdot \vec{v})(\vec{x},t)\, dV_x.$$

The details are left for an exercise. ///